

# RATNESH KUMAR JAISWAL

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## Education

### National Institute of Technology Delhi

08/2023 – 05/2027

Bachelor of Technology, Computer Science Engineering (GPA: 7.2)

New Delhi, India

## Experience

### Microsoft IDC

May 2026 – July 2026

Software Development Engineer Intern, Content Store (Agentic Security)

Bengaluru, India

- Designed and shipped an **automated Azure EV2 rollout-generation framework**, converting manual creation of **15+ deployment artifacts** per workload across **Full Forest** and **Ring-based deployments** into **1 deployment**.
- Built a **workload-scoped Service Model generator** that **reduced deployment artifact size by 98% (3,375 to 54 resources)**, improving validation efficiency and reducing new-workload onboarding time from **2 weeks to 2 days**.
- Developed **77 automated unit tests** with **mutation testing** as a **build-blocking quality gate** and delivered the solution through **5 merged production pull requests**.

### Indian Institute of Technology Mandi

June 2025 – July 2025

Research Intern

Mandi, Himachal Pradesh

- Developed an **Autonomous Ground Vehicle (AGV) detection system** using real-time drone video with **Detectron2** and a custom **CSPDarkNet-53 backbone** for enhanced feature extraction.
- Achieved **10% lower latency** with only **3% reduced mAP** compared to stable detection baselines.
- Outperformed **YOLO**, **Mask-RCNN**, default **Detectron2**, and **RT-DETR** in real-time drone-based AGV detection tasks.

## Projects

### Renergy - Walmart Sparkathon 2025 | Scikit-learn, XGBoost, AWS, Next.js, FastAPI

06/2025

- Engineered a **Next.js dashboard** with a **FastAPI backend**, delivering **real-time energy insights** to managers for informed, sustainable renewable-energy decisions.
- Developed a **real-time data pipeline** on **AWS S3** using **AWS Lambda** and **EventBridge** for continuous data integration, enabling predictive modeling for optimized energy use across Walmart.
- Built ML models (**XGBoost forecasting with 95–98% accuracy** and a rule-based optimization algorithm) that significantly **reduce costs and carbon footprint**.

### Imagin3D | Hugging Face, PyTorch, Flask, Transformers

03/2025

- Designed a **3D image reconstruction model** using a **ViT-base-patch16-224 Transformer** trained on the **Pix3D dataset** for high-quality multi-view reconstruction.
- Deployed the model on **Hugging Face** for accessible inference.
- Built a **Flask API** and deployed the web app on **Render**, enabling users to upload multiple 2D images and generate **real-time 3D models** for preview and download.

## Technical Skills

**Languages:** Python, Java, C, SQL

**Technologies:** Machine Learning, Deep Learning, DBMS, AWS, Docker, Agentic AI, RAG, Azure DevOps, CI/CD

**Libraries/Frameworks:** TensorFlow, PyTorch, Scikit-learn, Flask, FastAPI, Tailwind CSS, Next.js, LangChain

## Achievements

- Selected for **Amazon ML Summer School 2025**, gaining expertise in deep learning architectures, reinforcement learning, and large-scale ML systems.
- **Patent** on a Wall-Climbing and Cleaning Robot and an ongoing **research paper** on 3D image reconstruction.
- **Global Rank 6** at the **Technoxian World Innovation Challenge** with a prototype ADAS system for closed industrial environments.
- **Deputy General Secretary** of Kinetic Robotics Club, NIT Delhi, and member of the NIT Delhi Cricket Team.